

Jerry Yan

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EDUCATION

Purdue University

2025 - present PhD (Computer Science)

Advisor: Prof. Sotiris Nousias

UC San Diego

2023 - 2025 MS (Machine Learning & Data Science)

(GPA: 3.677)

Co-advisors: Prof. Nicholas Antipa and Prof. Sophia Merrifield

2019 - 2023 BS (Probability & Statistics)

(GPA: 3.914)

PUBLICATIONS

Yan, Jerry*, Matteo Forlivesi*, Bowen Tan, Andrew Xie, Siddharth Somasundaram, and Sotiris Nousias (2026). "Spatiotemporal Flux Probing for Single-Photon Videography". In: *Proceedings of European Conference on Computer Vision (ECCV)*. Springer.

Yan, Jerry, Chinmay Talegaonkar, Nicholas Antipa, Eric Terrill, and Sophia Merrifield (2026). "PoseIDON: 6DoF pose estimation with foundation model features for marine sediment burial mapping". In: *ISPRS Journal of Photogrammetry and Remote Sensing* 239, pp. 928–942. DOI: [10.1016/j.isprsjprs.2026.06.032](https://doi.org/10.1016/j.isprsjprs.2026.06.032).

Yan, Jerry, Thomas Cook, Allison Ho, Mark Otero, Peter A. Rogowski, Mika Siegelman, Eric Terrill, and Sophia Merrifield (2023). "PyPlume: A toolkit for rapid ocean surface transport assessments". In: *Environmental Modelling & Software* 168, p. 105783. ISSN: 1364-8152. DOI: [10.1016/j.envsoft.2023.105783](https://doi.org/10.1016/j.envsoft.2023.105783).

*Denotes equal contribution.

RESEARCH EXPERIENCE

Graduate Researcher · UCSD Computational Imaging Systems Lab

June 2023 – June 2025 · La Jolla, CA · (Principal Investigator: Nicholas Antipa, PhD)

- Developed methods to map burial states of dumped seabed waste from underwater vehicle imagery.
- Performed pose estimation of buried, degraded objects by leveraging foundation model features.
- Implemented buried barrel pose estimation via multiview 3D reconstruction and point cloud analysis.

Undergraduate Researcher · Coastal Observing Research & Development Center

April 2020 – June 2023 · La Jolla, CA · (Principal Investigator: Sophia Merrifield, PhD)

- Analyzed video and sidescan sonar images from underwater robotic vehicles to map ocean dumpsites.
- Created Python framework to run Lagrangian particle simulations on ocean models. This framework was used with HF radar data to help scientists respond to the Huntington oil spill.
- Processed and analyzed X-band radar data for western pacific maritime domain awareness.

EXPERIENCE

Computer Vision Programmer · Triton Unmanned Aerial Systems

October 2019 – June 2023 · La Jolla, CA · (5th Place Internationally in the 2022 AUVSI SUAS Competition)

- Created and programmed drones to compete in the Student Unmanned Aerial Systems Competition.
- Built and tested traditional computer vision and neural network models (such as U-Net) to perform image classification and segmentation on aerial images.
- Designed standardized testing frameworks for computer vision pipeline.
- Onboarded and mentored new software members through team projects.

Software Development Engineer Intern · Amazon Web Services

June 2022 – September 2022 · Seattle, WA

- Deployed new debugging features to improve troubleshooting capabilities for Amazon Just Walk Out services, which impact Amazon Go and Amazon Fresh stores.
- Communicated actively between different teams to determine implementation details and manage project pipelines.

SKILLS

Programming languages	Python, MATLAB, R, Java, Kotlin, C#, JavaScript
Tools & Libraries	PyTorch, OpenCV, Unity, Blender, Pandas